DEMO / SYNTHETIC: Influence of layer height and build orientation on FDM PLA tensile strength

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Abstract

DEMO / SYNTHETIC TEXT generated to test the AM-KG pipeline; the numbers are not real research results. PLA tensile specimens were printed by fused deposition modeling (FDM) with an extrusion temperature of 210 °C and a layer thickness of 0.2 mm. The build orientation affected the tensile strength, and a larger layer thickness increased delamination.

2. Materials and methods

Dog-bone specimens of PLA were printed by FDM with a nozzle diameter of 0.4 mm, a printing speed of 50 mm/s, a bed temperature of 60 °C and an infill density of 100 %. The layer thickness was varied from 0.1 mm to 0.3 mm. Tensile tests were performed according to a standard test method.

3. Results

The tensile strength of flat specimens was 58 MPa, whereas upright specimens reached only 31 MPa. A higher extrusion temperature improved interlayer bonding and reduced delamination. Increasing layer thickness decreased the tensile strength.

4. Conclusion

Build orientation governed the tensile strength of FDM PLA parts.